

SHOGUN User Manual

Drum machine with patch bay · Jidai Collection · Martial Systems

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1. Overview

SHOGUN is an analog-style drum machine with sixteen voices, a step sequencer for every voice and a patch bay. It has fourteen drum voices and two synth voices:

Voice	Sound
BD1, BD2	Two kicks. BD1 is a tuned resonator with pitch sweep, attack click, noise, filter and drive. BD2 is an oscillator kick with a tone control.
SD, RS	Snare (two tuned tones plus snappy noise) and rim shot.
CP, CL	Hand clap (1 to 8 bursts) and claves.
MA, CB	Maracas and cowbell.
CH, OH, CY	Closed hat, open hat and cymbal. The closed hat always chokes the open hat.
LTC, MTC, HTC	Low, mid and high toms. Each switches to a conga.
LEAD, BASS	Two mono synths with saw or square waves, resonant filter, envelope, glide and accent. LEAD has a state-variable filter, BASS a 4-pole ladder filter.

Around the voices:

- **A sequencer with a track per voice.** Each track has its own length (1 to 32 steps), step scale, swing and timing shift, so tracks can run in polymeter. Steps carry accent (3 levels), flam, ratchet, probability, micro-timing, pitch bend and, on the synth tracks, a note and a tie. Patterns can also lock any parameter on a step.
- **A triple wave shaper** on BD1, BD2 and the three toms: three folding stages in series under one WAVE knob, with SHAPE, symmetry and audio-rate modulation.
- **Four LFOs and a 32-slot mod matrix**, with sources such as each voice's envelope, velocity, accent and per-hit random, plus the mod wheel and aftertouch.
- **A mixer** with pan, mute, solo and a delay send per voice, four buses with drive, tone and compressor, eight stereo aux outputs and a master section with drive, glue, width and a clipper.

- **A patch bay** of 151 jacks that follows the Jidai Cable Standard, so SHOGUN cables straight into BUSHIDO, RONIN and ORIGAMI in the JIDAI RACK.
- **Analog behaviour:** slow pitch drift while playing, and fixed per-voice tolerances that belong to the unit and are saved with the patch.

SHOGUN runs as a VST3 instrument and as a device in the JIDAI RACK. A new instance starts on the INIT kit with an empty pattern. The factory bank adds 21 kits, each with its own pattern.

2. Quick start

1. Insert SHOGUN on an instrument track in your DAW (section 7).
2. Click the  key next to KIT and pick a kit, for example **002 Deep Round Kick House**. The kit and its pattern load together.
3. Choose a clock with the **SRC** key in the header: - **HOST** plays while your DAW plays, locked to the song position. Press play in the DAW. With the DAW stopped,  **PLAY** previews the pattern at the DAW's tempo until the DAW starts. - **INT** runs on SHOGUN's own TEMPO. Click  **PLAY** to start and stop, or press the space bar while the SHOGUN window has focus.
4. Turn the voice knobs on the **MAIN** tab. Click a voice name (BD1, SD and so on) to select it and hear it.
5. Edit the selected voice's steps in the STEPS row: click a step to turn it on or off, shift-click for an accent.
6. Open **GRID** to see and edit all sixteen tracks at once.
7. Play SHOGUN from a MIDI keyboard: drums on notes 36 to 49, LEAD on channel 1, BASS on channel 2 (section 5).

SRC belongs to the SHOGUN instance, not to the kit: loading a kit or INIT PATCH keeps HOST, INT or EXT as you set it. Your DAW project saves and restores it with the rest of the plugin's state.

3. Panel reference

The panel is 1200 × 672 points with eight tabs: MAIN, VOICE, GRID, MOD, ROUTE, FX/MIX, SEQ/MIDI and GLOBAL. The header and the CLOCK, SYNC and MASTER strip stay in view on every tab.

3.1 Using the controls

Action	Result
Drag a knob up or down	Changes its value. Hold Shift for fine control (4× finer).
Mouse wheel over a knob	Small steps.
Double-click a knob	Returns it to its default (the INIT kit value).
Click a key or display	Steps to the next choice. Toggles flip. Lists of 10 or more choices open as a menu instead.
Shift-click a key or display	Steps back one choice.
Right-click a key or display	Opens the full list as a menu, with the current choice ticked.
Click a knob's value readout	Opens a menu of values (TEMPO and A4 have their own presets and a typed entry).

Action	Result
Cmd-Z / Ctrl-Z	Undo. Cmd-Shift-Z or Cmd-Y (Ctrl-Shift-Z or Ctrl-Y) redo. These work while the SHOGUN window has focus; some DAWs keep these keys for themselves, and the ↶ ↷ keys always work.
Space bar	Starts and stops ► PLAY while the SHOGUN window has focus. While the DAW's transport plays, the space bar goes to the DAW.
Click a tab	Shows that tab.

Every sound, mixer, clock and global control is a plugin parameter (454 in all), so your DAW can automate it. Knob moves, step edits, cable changes and kit loads can all be undone.

3.2 Header

Control	What it does
Tabs	MAIN, VOICE, GRID, MOD, ROUTE, FX/MIX, SEQ/MIDI, GLOBAL.
KIT display	The loaded program's name. Click it for the program list.
KIT ◀ ▶	Steps to the previous or next program: INIT, then the 21 factory kits. The list wraps around.
PATTERN display	The loaded pattern's name. Click it for the program list.
PATTERN ◀ ▶	The same program steps as the KIT arrows. Every program is a kit with its matching pattern.
⌘	Opens the program list.
A / B	Compare two versions of the whole patch. Click a slot to switch to it (the first visit to B starts as a copy of A). Right-click or shift-click a slot to copy the other slot onto it.
↶ / ↷	Undo and redo, 64 levels. Also Cmd-Z / Ctrl-Z and Cmd-Shift-Z / Ctrl-Y (section 3.1).

3.3 CLOCK, SYNC and MASTER strip

Control	Range	What it does
► PLAY		In the TRANSPORT group. INT: starts and stops the sequencer. HOST with the DAW stopped: plays a preview at the DAW's tempo, and the DAW takes over when it starts. EXT: arms the sequencer for CLK IN. The light next to it shows that it is running. The space bar does the same while the SHOGUN window has focus.
RST		Restarts the pattern from step 1.
INT / EXT switch	INT, EXT	Where the voices are played from. INT : the pattern plays the voices. EXT : only the TRIG and GATE jacks play them (section 4).
TEMPO	40 to 200 BPM, default 120	SHOGUN's own tempo. With SRC on HOST, the host tempo is used. Click the readout for a menu of tempos, Type tempo... and Sync to DAW . Picking a tempo while SRC is HOST switches SRC to INT.
SWING	50 to 75 %, default 50 %	Delays every second step. 50 % is straight.
SCALE	1/32, 1/16, 1/8T, 1/8	Step length for tracks set to GLOBAL. Default 1/16.
BAR	1 to 32 steps, default 16	Bar length in steps. Sets the STEP/BAR count, the RST OUT pulse and LFO bar restarts.
STEP / BAR		Bar and step position, with a run light.

Control	Range	What it does
CLK IN display	STEP, 1, 2, 4, 24, 48 PPQN	How SHOGUN reads pulses at CLK IN when SRC is EXT: one pulse per step, or 1 to 48 pulses per quarter note.
SRC	HOST, INT, EXT	The clock source (section 5). Click for the next, shift-click for the previous, right-click for the list.
ACCENT	0 to 100 %, default 100 %	How much velocity and accent change the level. At 0 every hit plays at full level.
DRIVE	OFF to +24 dB	Master drive.
VOLUME	-inf to +6 dB, default 0 dB	Master volume.
Meter, CLIP , CPU		Output level (-48 to 0 dBFS), a clip light (click it to clear) and the processing load.

3.4 MAIN tab



VOICES. Three knobs per voice, with the voice's name key on top. Click a name key to select the voice and audition it. The red light under each name key flashes when the voice plays. The knobs are each voice's most used controls:

Voice	Knobs
BD1, BD2, SD, CL, CB, CH, CY, LTC, MTC, HTC	TUNE, DECAY, LEVEL (SD: TUNE, T.DECAY, LEVEL)
RS	TUNE, PAN, LEVEL
CP	FILTER, DECAY, LEVEL
MA	PAN, DECAY, LEVEL
OH	HAT TUNE (shared with CH), DECAY, LEVEL
LEAD, BASS	CUTOFF, RESO, LEVEL

STEPS. The selected voice's track, 16 steps at a time.

Control	What it does
Step keys	Click: step on or off. Shift-click: step on and toggle its accent. Right-click: select the step without changing it. Clicking a step past the track's length turns it on and extends the length to that step.
Track display	The selected track and its length.
1-16 / 17-32	Shows the first or second half of a 32-step track.
LEN display	The track's length and scale (GLOBAL or its own).
◀ TRK / TRK ▶	Selects the previous or next track.

STEP · LOCKS. The selected step's settings:

Control	Range	What it does
ACCENT	3 levels	Soft, normal or accented.
FLAM	OFF, 1 to 16	Plays the hit 2 to 5 times in quick succession, with gaps from 3.75 to 15 ms. Not used on CP.
RATCHET	1, 2, 3, 4, 6, 8	Splits the step into that many even hits.
PROB	0 to 100 %	Chance that the step plays. The pattern's random seed fixes each result, so the song plays the same way every time from the same point.
MICRO	± half a step	Moves the hit early or late.
BEND	-12 to +12 semitones	Pitch bend for the hit.
NOTE	C1 to C7	The note for a LEAD or BASS step.
COPY / PASTE		Copies the selected track's steps and length, and pastes them onto another track.
CLEAR		Clears the selected track.
RANDOM		Turns on a random 30 % of the track's steps.

3.5 VOICE tab



Every sound control of every voice, in voice groups:

Group	Controls and ranges
BD 1	TUNE 35 to 140 Hz · PITCH (sweep depth) 0 to +18 semitones · DECAY 8 to 720 ms · ATTACK (click) · SOUND (attack colour, CLN and 1 to 15) · NOISE · FILTER 200 Hz to 8 kHz · DRIVE OFF to 9 · WAVE · LEVEL
BD 2	TUNE 45 to 100 Hz · DECAY · TONE · WAVE · LEVEL
SNARE · RIM	TUNE 120 to 400 Hz · DETUNE ±8 semitones (second tone) · PITCH (sweep) 0 to +14 semitones · T.DECAY (tone decay) · TONE · SNAPPY · SN.DEC (snappy decay) · LEVEL · RIM TUNE 250 Hz to 2.5 kHz · RIM LVL
CLAP · CLAVES	ATTACK · SOUND (burst colour, CLN and 1 to 15) · COUNT 1 to 8 bursts · FILTER 400 Hz to 6 kHz · DECAY · LEVEL · CL TUNE 400 Hz to 3 kHz · CL DECAY · CL LEVEL
CB · MA	CB TUNE 300 Hz to 1.2 kHz · CB DECAY · CB LEVEL · MA DECAY · MA LEVEL
HATS · CYMBAL	HAT TUNE ×0.5 to ×2 · CH DECAY · OH DECAY · CH LEVEL · OH LEVEL · CY TUNE ×0.5 to ×2 · CY TONE · CY DECAY · CY LEVEL
TOMS · CONGAS	Per tom: TUNE (LTC 70 to 180 Hz, MTC 100 to 280 Hz, HTC 140 to 400 Hz) · DECAY · WAVE · LEVEL · TOM/CGA switch · NZ (noise on) · and the shared TOM NZ level
LEAD, BASS	CUTOFF (LEAD 200 Hz to 8 kHz, BASS 80 Hz to 4 kHz) · RESO · ENV (filter envelope depth) · DECAY · SAW/SQR · OCT -1, 0, +1 · TUNE ±100 cents · GLIDE 2 to 489 ms · ACCENT · LEVEL

Every DECAY runs from 8 to 720 ms. Click a group's name to select its voice. The light in a group's corner flashes red when its voice plays. Click the light to mute or unmute the voice: it stays green while the voice is muted, and it follows the **M** keys on GRID and FX/MIX.

SELECTED VOICE. Settings for the selected voice:

Control	Range	What it does
VEL → LEVEL	0 to 100 %, default 100 %	How much velocity and accent change the voice's level.

Control	Range	What it does
VEL → DECAY	-100 to +100 %	Velocity lengthens or shortens the decay.
ACC AMT	0 to 100 %, default 100 %	Accent depth for the voice.
PAN	-100 to +100 %	Equal-power pan.
CHOKE	OFF, 1 to 4	Voices in the same group cut each other off.
CV AMT	-100 to +100 %, default +100 %	Drum voices: scales the PITCH, DECAY and TONE jacks. LEAD and BASS: scales the pitch CV at NOTE and V/OCT. +100 % tracks exactly 1 V/oct, +50 % tracks half (1 V moves 6 semitones), 0 % holds NOTE at C3, negative values invert around C3.

WAVE · TRIPLE WAVE SHAPER. For the selected kick or tom (BD1, BD2, LTC, MTC, HTC). Three folding stages run in series. The front WAVE knob (on the voice) is a macro that brings in stage 1, then 2, then 3.

Control	Range	What it does
SHAPE	SIN, TRI, SAW	Morphs the voice's body from sine through triangle to saw before the shaper.
POST / PRE-VCA		POST shapes the finished voice. PRE-VCA shapes it before the envelope, so the envelope drives the folds.
WAVE 1, 2, 3	-100 to +100 %	Per-stage trims on top of the macro.
SYM 1, 2, 3	-100 to +100 %	Tilts each stage's fold for even harmonics.
VC → AMT, VC → SYM	-100 to +100 %	How much the VC source moves each stage's amount and symmetry.
VC SOURCE	BODY, NOISE, LFO 1 to 4, any voice	The internal modulation source, at audio rate. A cable into the voice's FOLD VC jack replaces it.
LEVEL COMP	OFF, ON	Keeps the level steady while you turn WAVE up.

With WAVE and the WAVE 1 to 3 trims at 0, the shaper is bypassed, whatever the SYM settings: the sound passes through untouched. A stage whose amount is 0 is a straight wire at any SYM, so SYM only colours a stage that WAVE or its trim has opened. VC → AMT with a live VC source and SHAPE above 0 take the shaper out of bypass.

3.6 GRID tab

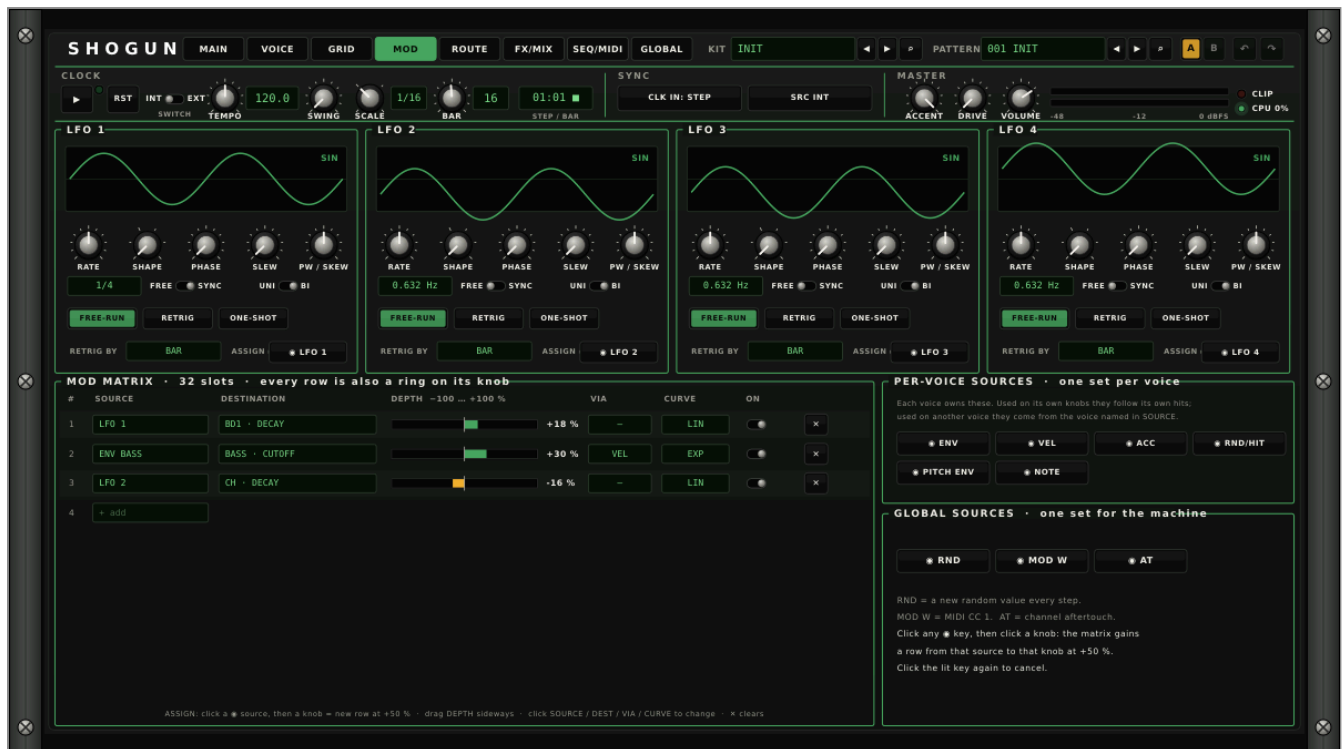


All sixteen tracks, 32 steps wide.

Control	What it does
Grid cell	Click: off → on → accent → off. Shift-click or right-click: select the step and its track.
M / S	Mute and solo for the track.
LEN	The track's length. Click for a menu of lengths 1 to 32 (the current one ticked), Type length... and Apply to all tracks . Shift-click removes a step.
Step knobs	ACCENT, FLAM, RATCHET, PROB, MICRO, BEND and NOTE for the selected step, as on MAIN.
TIE	On a LEAD or BASS step, holds the previous note into this step (legato with glide).
LOCKS ×	Clears the selected step's parameter locks. The display shows how many locks the step has and whether it is tied.
COPY / PASTE / CLEAR	As on MAIN, for the selected track.
SHIFT ◀ / SHIFT ▶	Rotates the selected track one step left or right.
RANDOM	Turns on a random 30 % of the selected track's steps.
DRAG MIDI	Drag it to your DAW or desktop to export the pattern as a MIDI file: one full cycle of every track, drums on channel 10 (notes 36 to 49), LEAD on channel 1 and BASS on channel 2, velocity from the accent, ratchets and flams as repeated notes. Probability isn't written: every step that is on is in the file.

Parameter locks come with the factory patterns and with saved patches. A locked step plays its own value for that parameter.

3.7 MOD tab



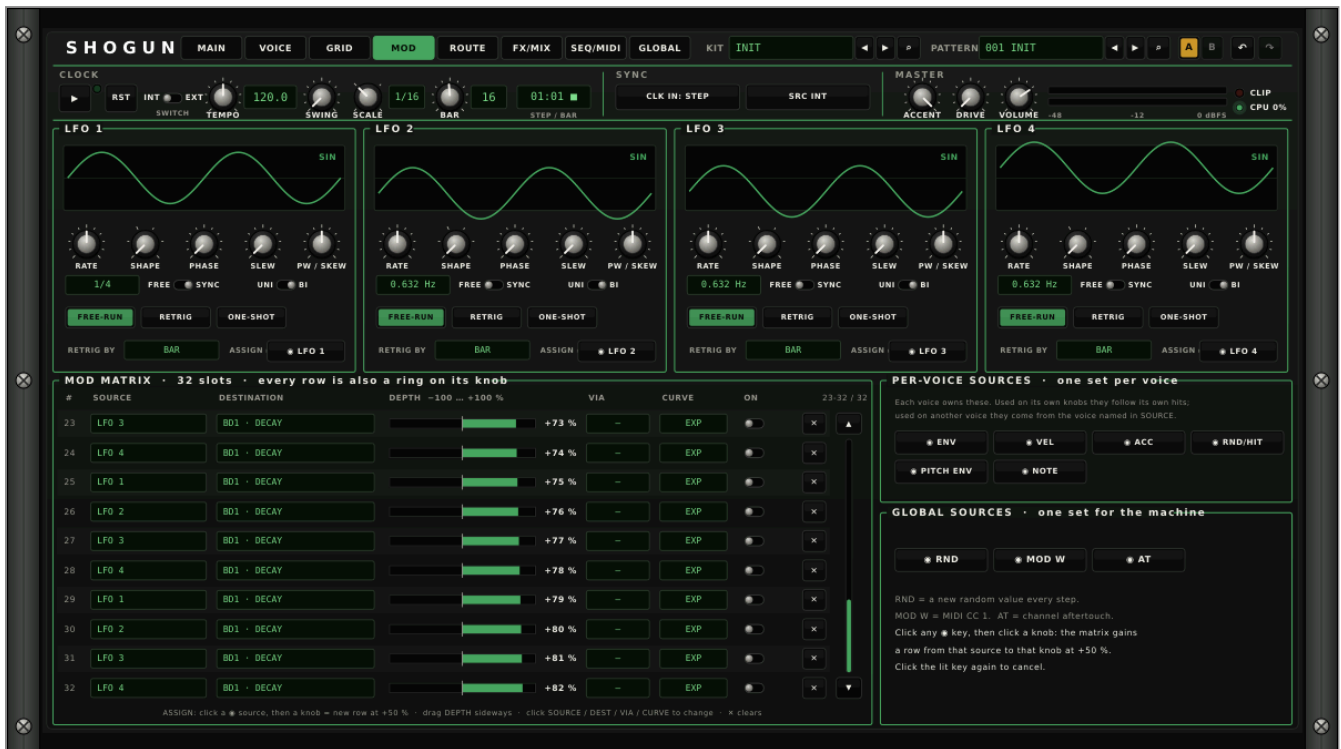
LFO 1 to 4. Each LFO shows its waveform.

Control	Range	What it does
RATE	0.01 to 40 Hz (FREE) or 8 BAR to 1/64T (SYNC)	Speed. The display shows Hz or the note value.
SHAPE	SIN, TRI, RAMP, SAW, SQR, S&H	Waveform.
PHASE	0 to 360°	Start phase.
SLEW	0 to 100 %	Smooths steps, for example on S&H.
PW / SKEW	0 to 100 %	Pulse width or skew.
FREE / SYNC		Free rate in Hz, or locked to the tempo.
UNI / BI		Unipolar or bipolar output.
FREE-RUN / RETRIG / ONE-SHOT		Runs freely, restarts on each RETRIG event, or runs once per event.
RETRIG BY	BAR, ANY TRIG, OWN VOICE, one voice's TRIG	What restarts the LFO. OWN VOICE gives every voice its own copy of the LFO, restarted by its own hits.
ASSIGN		Arms this LFO as a source for one-click assignment (below).

MOD MATRIX. 32 slots. Each row has a SOURCE, a DESTINATION (any continuous parameter), a DEPTH from -100 to +100 %, an optional VIA source that scales the depth, a CURVE (LIN, EXP, LOG, S-CRV) and an ON switch.

- Click SOURCE, DEST or VIA for a menu. Per-voice sources offer OWN VOICE or a named voice.
- Drag DEPTH sideways.
- Click CURVE to step through the curves, ON to switch the row, **x** to clear it.
- Click **+ add** (the SOURCE of the empty row) to start a new row. It is always the last row in the list.

- Ten rows show at a time. With more in use, a scroll bar appears at the right edge with ▲ and ▼ keys: the keys move a page of ten rows, the mouse wheel over the matrix moves one row, and the header shows which rows are in view (for example 23-32 / 32). Every row up to all 32 can be reached, edited and cleared.
- Every row also shows as a ring on its destination knob.



PER-VOICE SOURCES (each voice has its own): **ENV**, **VEL**, **ACC**, **RND/HIT** (a new random value per hit), **PITCH ENV**, **NOTE**. **GLOBAL SOURCES**: **RND** (a new random value every step), **MOD W** (MIDI CC 1) and **AT** (channel aftertouch).

Assigning in one click. Click a ● source key (or an LFO's ASSIGN key), then click any knob: the matrix gains a row from that source to that knob at +50 %. Click the lit key again to cancel.

3.8 ROUTE tab



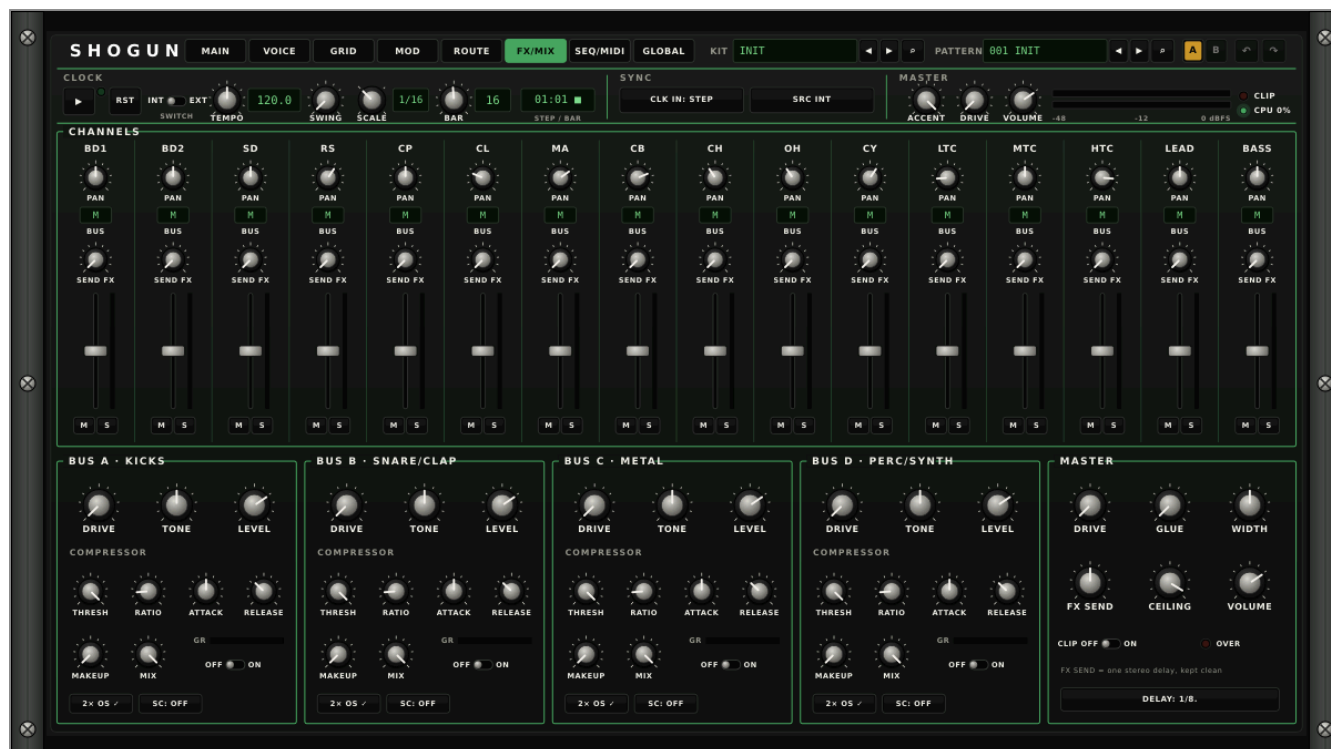
PATCH BAY. All of SHOGUN's jacks, one column per voice and a CLOCK · MOD · MIX block. Section 4 lists every jack.

- Drag from one jack to another to patch a cable.
- Right-click a jack to unplug its cables.
- Input labels are plain and output labels are reversed (light label). A ►SEQ mark shows that TRIG is normalled to the sequencer. Ring colours show the cable role.

PER-VOICE ROUTING. For every voice:

Control	Range	What it does
PAN	-100 to +100 %	Equal-power pan.
OUTPUT	MAIN, BUS A to D, AUX 1/2 to AUX 15/16, PAIR	Where the voice goes (section 7 lists the PAIR outputs).
CHOKE	OFF, 1 to 4	Choke group.
CV AMT	-100 to +100 %	Drum voices: PITCH, DECAY and TONE jacks. LEAD and BASS: the pitch CV at NOTE and V/OCT (+100 % = exact 1 V/oct).

3.9 FX/MIX tab



CHANNELS. One strip per voice: **PAN**, **BUS** (shows the voice's OUTPUT: M for MAIN, A to D, an aux pair or PR for PAIR), **SEND FX** (to the stereo delay), a level fader and **M/S** (mute, solo).

BUS A to D (labelled KICKS, SNARE/CLAP, METAL, PERC/SYNTH; any voice can use any bus):

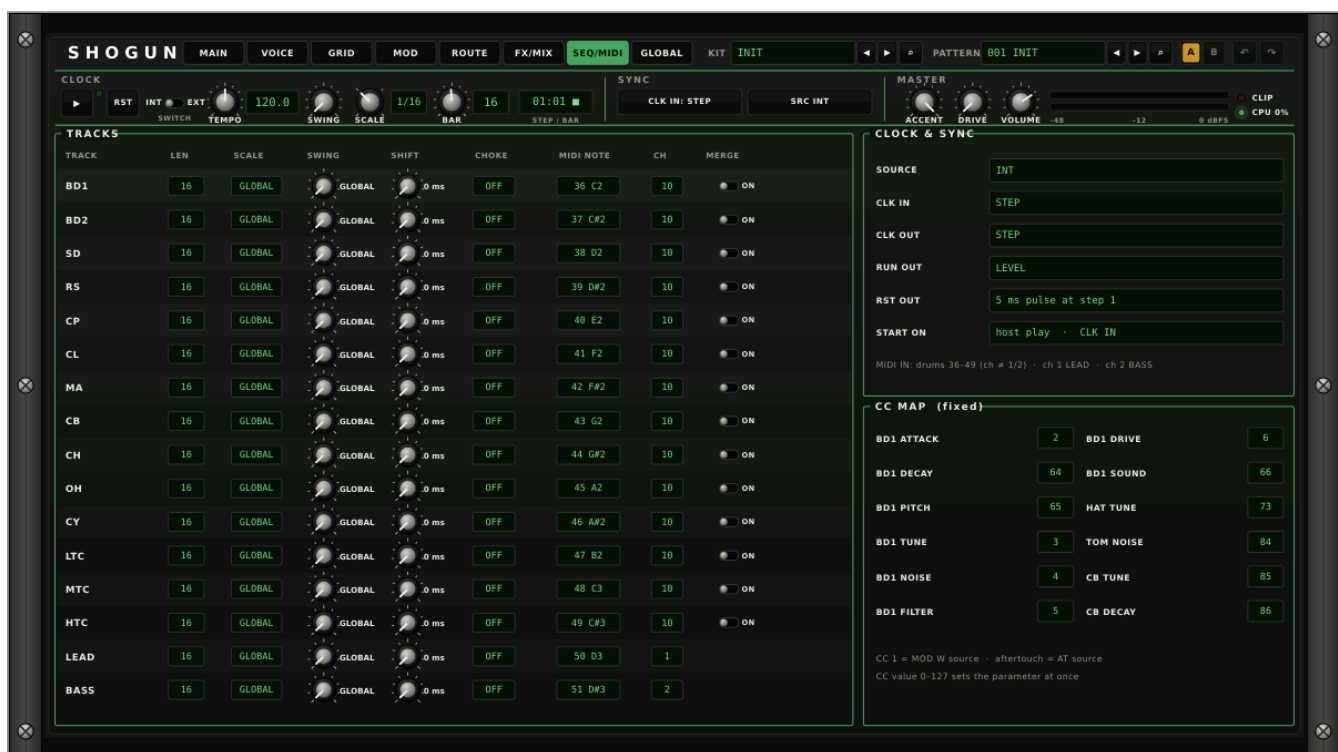
Control	Range	What it does
DRIVE	OFF to +24 dB	Bus saturation.
TONE	-100 to +100 %	Tilt EQ, darker to brighter.
LEVEL	-inf to +6 dB	Bus level into the master.
THRESH	-40 to 0 dB	Compressor threshold.
RATIO	1:1 to 20:1	Compressor ratio.
ATTACK	0.1 to 100 ms	Compressor attack.
RELEASE	10 to 1000 ms	Compressor release.
MAKEUP	0 to +24 dB	Makeup gain.
MIX	0 to 100 %	Parallel compression blend.
OFF / ON, GR		Compressor on, and its gain reduction meter.
SC	OFF or any voice	Sidechain: the compressor listens to that voice.
2x OS	1x, 2x, 4x	Shows the realtime oversampling. Click to change it (the same setting as on GLOBAL).

MASTER:

Control	Range	What it does
DRIVE	OFF to +24 dB	Master drive.

Control	Range	What it does
GLUE	0 to 100 %	Gentle bus compressor. 0 is off.
WIDTH	0 to 100 %, default 50 %	Stereo width. 0 is mono, 50 % unchanged, 100 % wider. Lows below 120 Hz are never widened.
FX SEND	0 to 100 %, default 50 %	Return level of the delay.
CEILING	-6 to 0 dBFS, default -0.3	Clipper ceiling.
VOLUME	-inf to +6 dB	Master volume.
CLIP OFF / ON, OVER		Soft clipper on the output, and a light when the output goes over.
DELAY	1/16 to 1/2T, default 1/8.	Delay time, synced to the tempo.

3.10 SEQ/MIDI tab



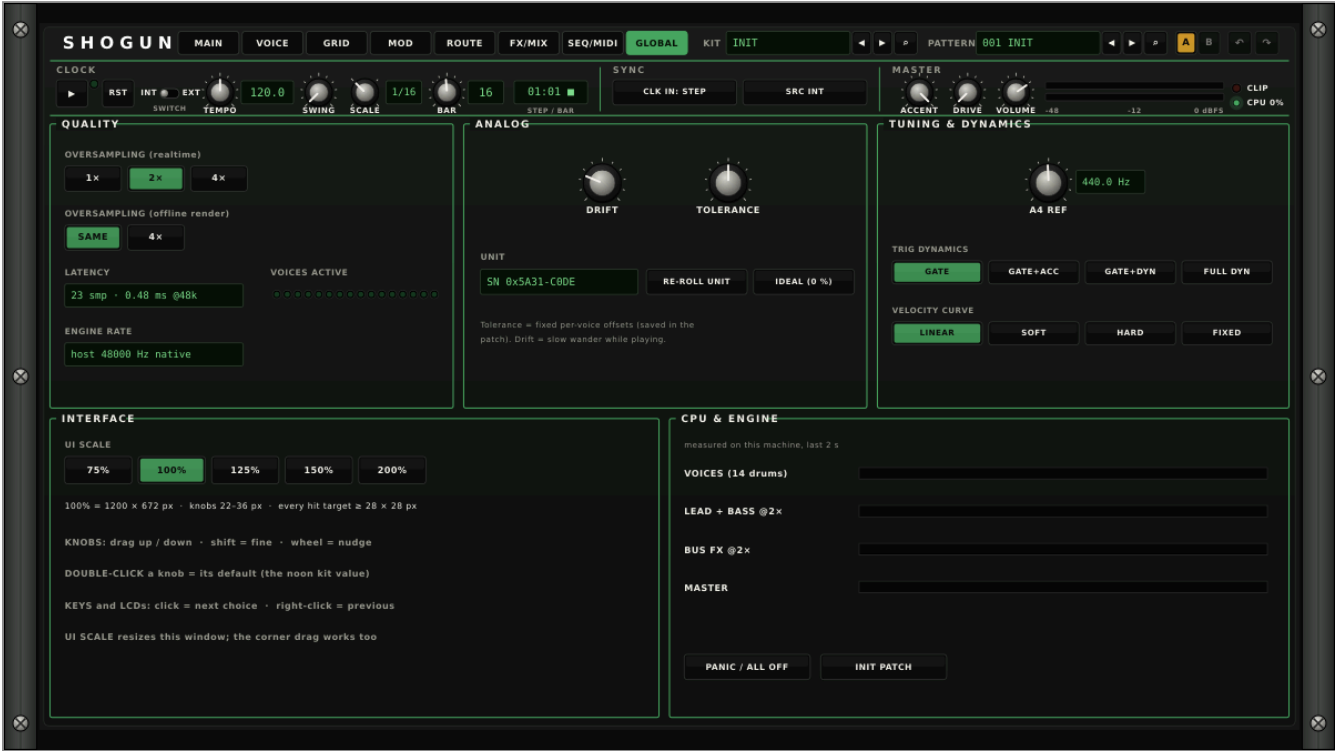
TRACKS. One row per track:

Column	What it does
LEN	Track length, 1 to 32.
SCALE	GLOBAL (follows the header SCALE), 1/32, 1/16, 1/8T or 1/8. Click to step.
SWING	GLOBAL (knob fully left) or the track's own swing, 50 to 75 %.
SHIFT	Delays the whole track by 0 to 30 ms.
CHOKE	Choke group, as on ROUTE.
MIDI NOTE, CH	The track's MIDI note and channel (section 5).
MERGE	Drum tracks: with the INT/EXT switch on INT, lets the TRIG jack play the voice alongside the pattern.

CLOCK & SYNC. SOURCE (HOST, INT, EXT), CLK IN (STEP or 1 to 48 PPQN), CLK OUT (STEP or 1 to 48 PPQN), RUN OUT (LEVEL or PULSE), RST OUT (5 ms pulse at step 1) and START ON (host play or CLK IN). Section 5 explains them.

CC MAP (fixed). The twelve controls on MIDI CCs (section 5).

3.11 GLOBAL tab



Section	Control	What it does
QUALITY	OVERSAMPLING (realtime) 1×, 2×, 4×	Default 2×. Higher settings reduce aliasing in drive and folding and add latency (section 8).
QUALITY	OVERSAMPLING (offline render) SAME, 4×	SAME uses the realtime setting when you export. 4× always renders at 4×.
QUALITY	LATENCY, VOICES ACTIVE, ENGINE RATE	The current latency in samples and ms, a light per sounding voice, and the host sample rate.
ANALOG	DRIFT 0 to 100 %, default 25 %	Slow pitch wander while playing.
ANALOG	TOLERANCE 0 to 100 %, default 50 %	Fixed per-voice offsets of pitch, decay and filter, as between real components.
ANALOG	UNIT, RE-ROLL UNIT	The unit's serial number, which fixes the tolerance offsets. RE-ROLL UNIT picks a new unit. The unit is saved with the patch.
ANALOG	IDEAL (0 %)	Sets DRIFT and TOLERANCE to 0.
TUNING & DYNAMICS	A4 REF 415 to 466 Hz	Tuning reference. Default 440 Hz.
TUNING & DYNAMICS	TRIG DYNAMICS	How the voltage of a TRIG jack sets the level when VEL is unpatched: GATE every hit full, GATE+ACC 4.5 V or more is an accent, GATE+DYN 2.5 to 5 V sets the level, FULL DYN 1 to 5 V sets the level.
TUNING & DYNAMICS	VELOCITY CURVE	MIDI velocity response: LINEAR, SOFT, HARD or FIXED (every note at full velocity).

Section	Control	What it does
INTERFACE	UI SCALE 75 %, 100 %, 125 %, 150 %, 200 %	Resizes the window. You can also drag its corner.
CPU & ENGINE	Meters	Load of the drum voices, LEAD and BASS, bus effects and master.
CPU & ENGINE	PANIC / ALL OFF	Stops the sequencer and restarts the pattern.
CPU & ENGINE	INIT PATCH	Loads INIT: default kit, empty pattern, no cables, no matrix rows. SRC stays as it is.

4. Patching

4.1 The patch bay

SHOGUN's patch bay is on the ROUTE tab. You can cable any output to any input, including SHOGUN to itself (a cable between two SHOGUN jacks has one sample of delay). Several cables into one input add up. In the JIDAI RACK the same jacks sit on SHOGUN's rear plate and patch to the other devices (appendix).

All jacks follow the **Jidai Cable Standard**:

Role	Colour	Level
Audio	Red	± 5 V is full scale
1 V/oct pitch	Blue	1 V per octave, 0 V = C3
Gate / clock	Green	Outputs 0 V or +5 V. Inputs go high above 1.0 V and low below 0.5 V.
CV	Yellow	± 5 V (or 0 to 5 V for unipolar sources)

The bay offers **151 jacks**. A patch keeps every cable when you save it, and patches made with older jack names still load with their cables in place. Early patches could hold a cable on **CLOCK > FILL IN** or **MOD > LANE A**, two jacks SHOGUN no longer has; such a patch still loads, without those cables, and the rest of it is untouched.

4.2 Normals and how inputs combine

A normalled input has an internal connection that a cable replaces:

Input	With no cable	With a cable
TRIG (drums), GATE (LEAD, BASS)	The pattern plays the voice (▶SEQ).	Rising edges play the voice when the INT/EXT switch is on EXT, or on INT with the track's MERGE on. On EXT the pattern no longer plays the voices. A synth note ends when its GATE falls.
VEL	Pattern hits use the step's accent level: 2.35 V (soft), 3.71 V (normal) or 5 V (accent). Jack hits use TRIG DYNAMICS.	The cable sets the velocity of every hit: 0 V quietest, 5 V full.
NOTE (LEAD, BASS)	The step's note.	The cable sets the note (1 V/oct, 0 V = C3) and moves a held note at once.
RET	The voice's own sound.	The cable replaces the voice's sound in its mixer channel. Pan, OUTPUT, SEND FX, mute and solo still apply.

Input	With no cable	With a cable
FOLD VC (BD1, BD2, toms)	The WAVE section's VC SOURCE.	The cable is the VC source, at audio rate. ± 5 V is full depth.

Modulation inputs add to their knob:

Input	Scaling
PITCH (drums)	1 V/oct on the voice's pitch, times CV AMT.
DECAY, TONE (drums)	± 5 V sweeps the knob's whole range, times CV AMT.
WAVE (BD1, BD2)	± 5 V sweeps the WAVE knob's whole range.
V/OCT (LEAD, BASS)	1 V/oct transposes the synth, times CV AMT.
CUTOFF (LEAD, BASS)	1 V/oct on the filter cutoff.

A voice's **OUT** jack carries a copy of the voice after its level and envelope, before pan. The voice stays in the mix, so to use an external effect as an insert, patch OUT to the effect and the effect back into the same voice's RET.

4.3 Jack table

Each drum voice (BD1, BD2, SD, RS, CP, CL, MA, CB, CH, OH, CY, LTC, MTC, HTC) has eight jacks:

Jack	Dir	Role	Signal
TRIG	in	Gate	A rising edge plays the voice.
VEL	in	CV	Velocity, 0 to 5 V.
PITCH	in	1 V/oct	Adds to the voice's pitch.
DECAY	in	CV	Adds to DECAY (± 5 V).
TONE	in	CV	Adds to the voice's tone control (± 5 V).
RET	in	Audio	Return into the voice's mixer channel.
OUT	out	Audio	The voice, post level and envelope, pre pan.
ENV	out	CV	The voice's envelope, 0 to 5 V, scaled by the hit's velocity.

BD1 and BD2 add **WAVE** (CV in). BD1, BD2, LTC, MTC and HTC add **FOLD VC** (CV in, audio rate).

LEAD and BASS each have eight jacks:

Jack	Dir	Role	Signal
GATE	in	Gate	Starts a note on the rising edge and ends it on the falling edge.
VEL	in	CV	Velocity, 0 to 5 V. 4.5 V or more is an accent.
NOTE	in	1 V/oct	The note, 0 V = C3, times the voice's CV AMT (+100 % = exact 1 V/oct).
V/OCT	in	1 V/oct	Transposes the synth, times the voice's CV AMT.
CUTOFF	in	CV	Adds to the cutoff, 1 V/oct.
RET	in	Audio	Return into the voice's mixer channel.
OUT	out	Audio	The voice, post level and envelope, pre pan.
NOTE OUT	out	1 V/oct	The note playing, including glide.

Clock, modulation and mix jacks:

Jack	Dir	Role	Signal
CLK IN	in	Gate	Clock pulses for SRC EXT (STEP or 1 to 48 PPQN, set by CLK IN).
RST IN	in	Gate	A rising edge restarts the pattern from step 1.
RUN IN	in	Gate	A rising edge starts or stops the sequencer.
CLK OUT	out	Gate	0/5 V square clock while running, at the step rate or 1 to 48 PPQN (set by CLK OUT).
RST OUT	out	Gate	5 ms, 5 V pulse at step 1 of every bar.
RUN OUT	out	Gate	LEVEL: 5 V while running. PULSE: a 5 ms pulse at start.
ACC OUT	out	CV	The accent level of the current step's drum hits (2.35, 3.71 or 5 V), 0 V on steps without a hit. Also active when the INT/EXT switch is on EXT.
LFO 1 to LFO 4	out	CV	The LFOs: 0 to 5 V (UNI) or ± 5 V (BI).
RND	out	CV	A new random value from 0 to 5 V on every step.
LD GATE, BS GATE	out	Gate	5 V while LEAD or BASS holds a note.
MIX L, MIX R	out	Audio	SHOGUN's main stereo mix.

The LFO and RND jacks copy sources that already work inside SHOGUN through the MOD tab without a cable. Use them to modulate other Jidai devices.

That is 112 drum jacks, 16 synth jacks and 23 clock, modulation and mix jacks: 151 in all.

4.4 Patch ideas

- **LFO 1 → BD1 PITCH**: a kick whose pitch moves with the LFO. Turn down BD1's CV AMT for a subtler move.
- **LEAD NOTE OUT → BASS NOTE**: BASS follows LEAD's notes. Program BASS steps where you want it to play.
- **CH ENV → BASS CUTOFF**: the hats open the bass filter on every hit.
- **CLK OUT → another device's clock**: drive BUSHIDO or any clock input from SHOGUN's tempo.
- **BD1 OUT → effect → BD1 RET**: an external effect inserted on the kick.

5. MIDI and host sync

5.1 Clock sources

The **SRC** key (header) and SOURCE (SEQ/MIDI) pick the clock:

SRC	Behaviour
HOST	Plays while the DAW plays, at the host tempo, locked to the song position. Loops and jumps land on the right step. With the DAW stopped, ► PLAY previews the pattern at the DAW's tempo until the DAW starts.
INT	Runs on TEMPO. Start and stop with ► or a pulse at RUN IN.
EXT	Steps on rising edges at CLK IN: one step per pulse with CLK IN on STEP, or 1 to 48 pulses per quarter note. Start it with ► or a pulse at RUN IN. RST IN restarts the pattern.

With SRC on HOST and no host transport (for example in a host that sends no play position), SHOGUN runs on its own TEMPO with ►.

SRC is a setting of the SHOGUN instance, as on the rack. Loading a kit (from the DAW's program list, the KIT and PATTERN displays or their arrows), INIT PATCH and switching between A and B keep the current SRC. Undo and redo include SRC changes, and the DAW project saves and restores SRC with

the plugin's state.

CLK OUT, RST OUT and RUN OUT follow whichever source is active, so SHOGUN can clock other devices in every mode.

5.2 Playing from the pattern or from jacks

The **INT/EXT** switch in the CLOCK strip sets what plays the voices. On **INT** the pattern plays them, and TRIG and GATE jacks are ignored unless that track's MERGE is on. On **EXT** only the TRIG and GATE jacks play them. The clock keeps running in both, so ACC OUT, the LFOs and the clock outputs carry on.

5.3 MIDI notes

Input	Plays
Notes 36 to 49 on any channel except 1 and 2	The drums: 36 BD1, 37 BD2, 38 SD, 39 RS, 40 CP, 41 CL, 42 MA, 43 CB, 44 CH, 45 OH, 46 CY, 47 LTC, 48 MTC, 49 HTC.
Notes 50 and 51 on those channels	LEAD and BASS at C3.
Any note on channel 1	LEAD at that note. MIDI note 48 is C3 (0 V).
Any note on channel 2	BASS at that note.

MIDI notes play the voices in both INT and EXT, and alongside the pattern. Velocity sets the level through the VELOCITY CURVE (GLOBAL).

MIDI out. Every step the pattern plays is also sent on SHOGUN's MIDI output at its exact sample position, with a note-off: drums on channel 10 (notes 36 to 49, as above), LEAD on channel 1 and BASS on channel 2 at the step's note, velocity 70, 100 or 127 from the accent.

5.4 MIDI CCs

Twelve voice controls have fixed CCs. A CC value from 0 to 127 sets the control at once over its whole range.

Control	CC	Control	CC
BD1 ATTACK	2	BD1 DRIVE	6
BD1 TUNE	3	BD1 DECAY	64
BD1 NOISE	4	BD1 PITCH	65
BD1 FILTER	5	BD1 SOUND	66
HAT TUNE	73	CB TUNE	85
TOM NOISE	84	CB DECAY	86

CC 1 (mod wheel) is the **MOD W** source and channel aftertouch is the **AT** source in the mod matrix.

6. Factory presets by bank

SHOGUN has one factory bank: INIT and 21 kits. Each kit loads with its own pattern, tempo and mod rows. Every kit is trimmed to peak at about -8 dBFS with the default master settings. With SRC on HOST, the DAW's tempo replaces the kit's tempo.

No.	Name	BPM	What it plays
001	INIT	120	The default kit with an empty pattern.
002	Deep Round Kick House	122	Long round kick, clap on two and four, off-beat open hats and a bass line with dotted-eighth delay, light swing.
003	Peak Time Warehouse	132	Driven kick, BD2 rumble and hats on a compressed bus keyed by the kick, glue and master drive.
004	Metallic Industrial	128	Folded kick, noisy folded toms through a driven bus, cowbell, cymbal and a ratcheted snare.
005	Electro Body Pop	126	Long kick, square-wave LEAD arpeggio and BASS, cowbell and claves.
006	Dusty Broken Beat	98	Swung broken beat with probability and micro-timing on snare and hats, high drift and tolerance.
007	Head Nod Boom Bap	90	Heavy swing, a laid-back snare and a long, low bass.
008	Minimal Click Groove	124	Short clicky kick, rim and claves with per-step tuning locks, and LFO 3 on the maraca decay.
009	Conga Groove	112	Toms switched to congas, claves, cowbell and maracas, with flams.
010	Half Time Pressure	140	Half-time snare, long BD2, a gliding bass and a hat ratchet roll.
011	Slow Ballad Brushes	72	12-step bar on 1/8T: soft kick and snare, a LEAD melody and a cymbal pattern with probability.
012	Seven Eight Stepper	118	14-step bar (7/8) with a square-wave bass.
013	Polymeter Drift	126	Tracks of 12, 10, 7, 5 and 9 steps against a four-on-the-floor kick, with four LFO rows.
014	Acid Line Workout	130	Resonant saw BASS with accents, slides and cutoff locks, swept by LFO 2.
015	Breakbeat Rush	136	Syncopated kick, ghost snares with a flam and a cymbal on the downbeat.
016	Two Step Shuffle	132	Strong swing, a skipping kick and a gliding bass.
017	Lead Arp Sequence	120	Sixteen-note saw LEAD arpeggio with octave locks, LFOs on cutoff and pan.
018	Dub Echo Chamber	118	Rim, clap and LEAD into a long dotted-eighth delay over a deep bass.
019	Rolling Hat Half Step	70	Hat ratchets up to 8, a long BD2 with pitch bends.
020	Shaker Percussion Circle	116	Maracas with micro-timing, cowbell, claves and toms.
021	Jungle Break Roller	170	Fast break with snare ratchets and a long gliding sub bass.
022	Lo-Fi Tape Wobble	84	LFOs wobble the LEAD and BASS tuning, with high drift and tolerance.

7. DAW setup

SHOGUN is a VST3 instrument with MIDI input and output, a stereo main output and eight optional stereo aux outputs.

Installing

Copy SHOGUN.vst3 into your system's VST3 folder:

System	VST3 folder
macOS	~/Library/Audio/Plug-Ins/VST3/ (just you) or /Library/Audio/Plug-Ins/VST3/ (all users)

System	VST3 folder
Windows	C:\Program Files\Common Files\VST3\
Linux	~/ .vst3/

Then have your DAW rescan its plugins. SHOGUN is listed under **Martial Systems**.

Using it in any DAW

1. Insert SHOGUN on an instrument track.
2. Set **SRC** to **HOST** so SHOGUN plays with your DAW's transport. Loading kits keeps it on HOST.
3. To play voices from MIDI, record or draw notes on the track (section 5).
4. Pick kits from your DAW's program list or from SHOGUN's KIT display. The DAW saves the whole patch with your project: kit, pattern, cables, matrix rows and unit.
5. Leave the DAW's plugin delay compensation on. SHOGUN reports its latency (0, 23 or 26 samples).
6. If your DAW doesn't show SHOGUN's MIDI output after an update, have it rescan its plugins.

Multiple outputs

Enable SHOGUN's aux outputs in your DAW to give voices their own mixer channels. A voice's OUTPUT setting (ROUTE tab) sends it to **AUX 1/2** (the plugin's Aux 1 output) through **AUX 15/16** (Aux 8). **PAIR** sends each voice to a fixed place:

Aux output	Left	Right
Aux 1	BD1	BD2
Aux 2	SD	RS
Aux 3	CH, OH	CY
Aux 4	CP (stereo)	
Aux 5	LTC, MTC, HTC (stereo)	
Aux 6	CL	CB
Aux 7	MA (stereo)	
Aux 8	LEAD	BASS

A voice routed to an aux output leaves the main mix. Its SEND FX still feeds the delay on the main mix.

Example: FL Studio

1. Close FL Studio, copy the VST3 into your VST3 folder, and open FL Studio.
2. Open **Options > Manage plugins** and click **Find installed plugins**.
3. Add SHOGUN to the Channel Rack and route it to a Mixer insert.
4. Set SRC to HOST and press play. SHOGUN follows the song position.
5. For multiple outputs, set the voices' OUTPUT to aux outputs, then route SHOGUN's outputs to Mixer inserts in the plugin wrapper's settings.
6. FL Studio's plugin delay compensation uses the latency SHOGUN reports.

8. Specs

Type	Drum machine: 14 drum voices and 2 synth voices, with a sequencer track per voice
Format	VST3 instrument. Also a device in the JIDAI RACK.

Outputs	Stereo main plus 8 optional stereo aux outputs
MIDI	Note input (drums on notes 36 to 49, LEAD channel 1, BASS channel 2), 12 fixed CCs, mod wheel, channel aftertouch. Note output of every played step. Pattern export as a MIDI file.
Sample rates	Runs at the host's sample rate. Tested at 44.1, 48 and 96 kHz.
Oversampling	1×, 2× (default) or 4× realtime. Offline render at the same setting or 4×.
Latency	0 samples at 1×, 23 at 2×, 26 at 4×, reported to the host (23 samples is 0.48 ms at 48 kHz)
Sequencer	16 tracks, 1 to 32 steps each, per-track scale, swing and shift, parameter locks
Modulation	4 LFOs, 32-slot matrix, per-voice and global sources
Mixer	4 buses with drive, tone and compressor, stereo delay send, master drive, glue, width and clipper
Patch bay	151 jacks under the Jidai Cable Standard
Parameters	454, all automatable
Factory programs	22 (INIT and 21 kits with patterns)
Undo	64 levels, plus A/B compare
UI	1200 × 672, 50 % to 200 % (keys for 75 % to 200 %)
Rack device	5.8 U open, 1 U closed, 4 U back. 0, 23 or 26 samples latency, compensated by the rack.

9. Troubleshooting

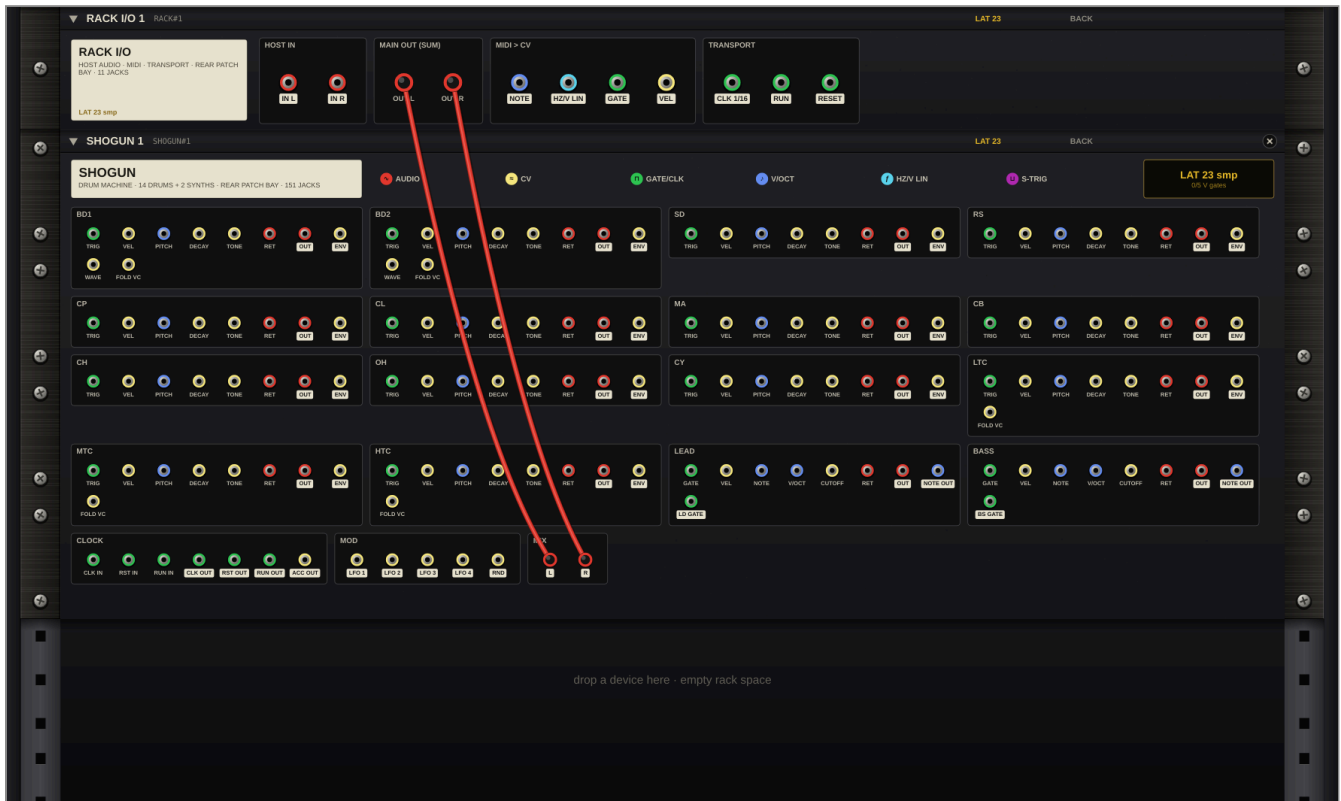
Problem	What to check
SHOGUN doesn't play with my DAW	Set SRC to HOST. Loading a kit keeps SRC as it is.
Nothing plays on INT	Click ► PLAY. Check that the INT/EXT switch is on INT.
Nothing plays on EXT	Click ► PLAY (or send a pulse to RUN IN), then send pulses to CLK IN. Check the CLK IN setting (STEP or PPQN).
The pattern runs but voices are silent	Check the INT/EXT switch: on EXT only TRIG and GATE jacks play the voices. Also check mute, solo and LEVEL.
A TRIG cable does nothing	On INT, turn on the track's MERGE (SEQ/MIDI), or switch to EXT.
A voice is missing from the main mix	Its OUTPUT is set to an aux output. Enable that output in your DAW, or set OUTPUT to MAIN.
A voice is silent with a cable in RET	RET replaces the voice's sound. Check the signal you return, or remove the RET cable.
Every hit plays at the same level	Check MASTER ACCENT and VEL → LEVEL, and VELOCITY CURVE (FIXED plays all notes at full velocity).
The drums are late against other tracks	Turn on plugin delay compensation in your DAW, or set OVERSAMPLING to 1×.
Clipping (CLIP lit)	Lower VOLUME or the voice levels, or turn on the CLIP soft clipper. Click CLIP to clear it.
Kit sounds slightly different each time on another computer	The unit (GLOBAL) sets the tolerances. It is saved with the patch. Click IDEAL for no tolerance or drift.
Plugin not listed	Check the VST3 folder (section 7) and rescan in your DAW.

10. Legal

SHOGUN, JIDAI RACK, BUSHIDO, RONIN, ORIGAMI and the Jidai Collection are products of Martial Systems LLC. All other trademarks belong to their owners. See the LICENSE file that comes with SHOGUN for the full terms.

Appendix: Back panel patching (SHOGUN in the JIDAI RACK)

In the JIDAI RACK, SHOGUN is a 5.8 U device when open and a 1 U strip when closed. Press **Tab** (or click **BACK** in the rack header) to flip the whole rack around. On the back, SHOGUN is a 4 U rear plate with a box for every voice and for CLOCK, MOD and MIX.



Rear jacks

The rear plate carries the same 151 jacks as the ROUTE tab (section 4.3), in the same colours.

Output jacks have a light label and inputs a plain one. The **LAT** plate shows SHOGUN's latency: 0, 23 or 26 samples at 1×, 2× or 4×.

Automatic cables

When you add a SHOGUN, the rack patches **MIX L/R** to **RACK I/O > MAIN OUT L/R**, and SHOGUN follows the host transport (SRC HOST). Press **K** to cycle the cable views: **ALL**, **HIDE PASS-THRU**, **SELECTED** and **HIDE**.

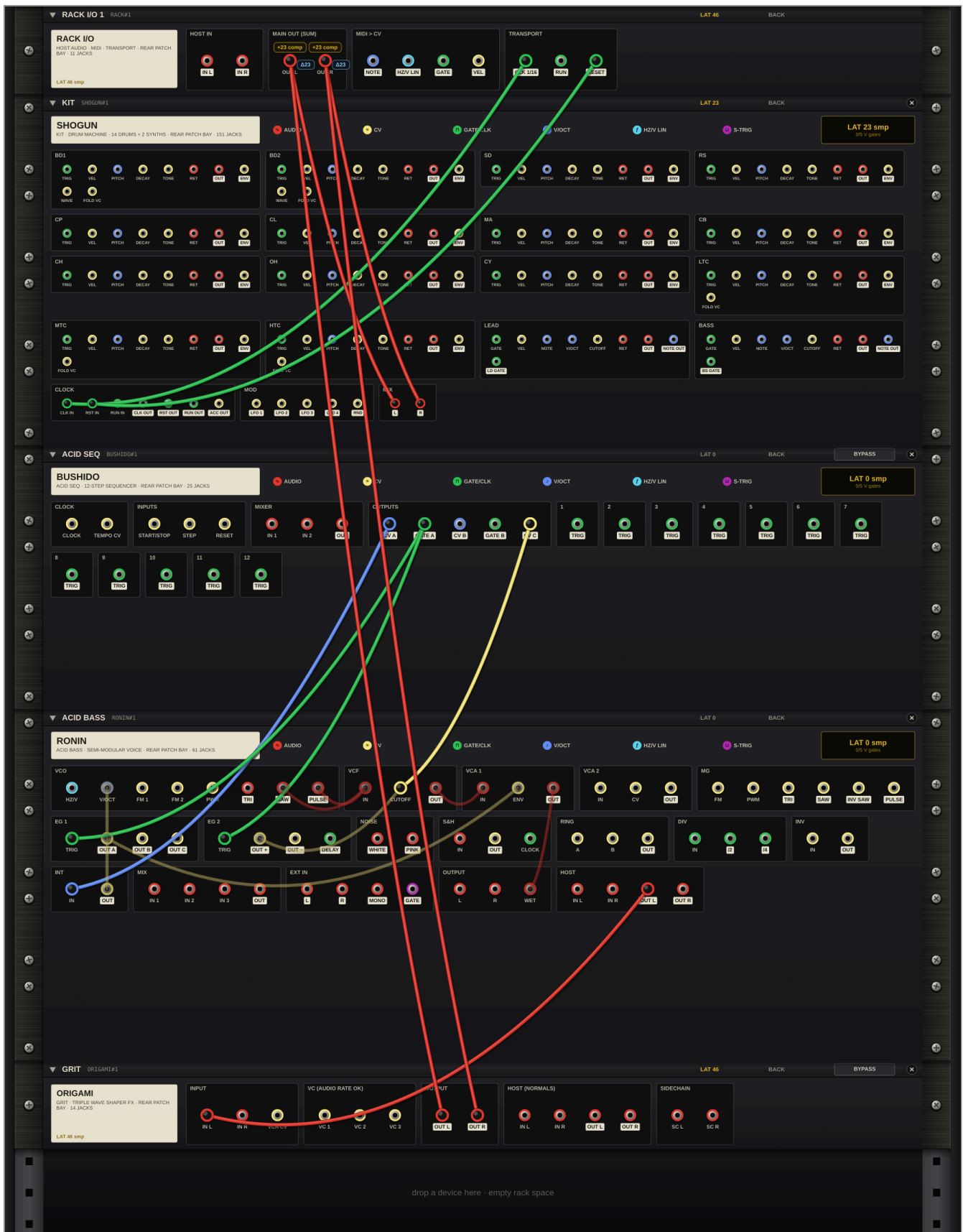
Patching other voices with SHOGUN

A SHOGUN **RET** jack feeds audio into SHOGUN's mix through its oversampled output stage, so at 2× or 4× a signal patched into RET reaches SHOGUN's MIX twice SHOGUN's delay later (46 or 52 samples), while SHOGUN's own voices arrive after the single delay (23 or 26). To keep an external voice tight with the drums, patch it to **MAIN OUT** next to SHOGUN's MIX, not into SHOGUN's RET, and let the rack line them up. The rack delays the faster paths into MAIN OUT to match the slowest and shows an amber **+n comp** tag on the delayed cables.

Use SHOGUN's RET jacks for effects you insert on a SHOGUN voice (OUT to the effect, the effect back into RET).

Worked example: Acid Drum Jam (starter rack)

Open the rack's **RACKS** menu and choose **EDM > Acid Drum Jam**, then press Tab.

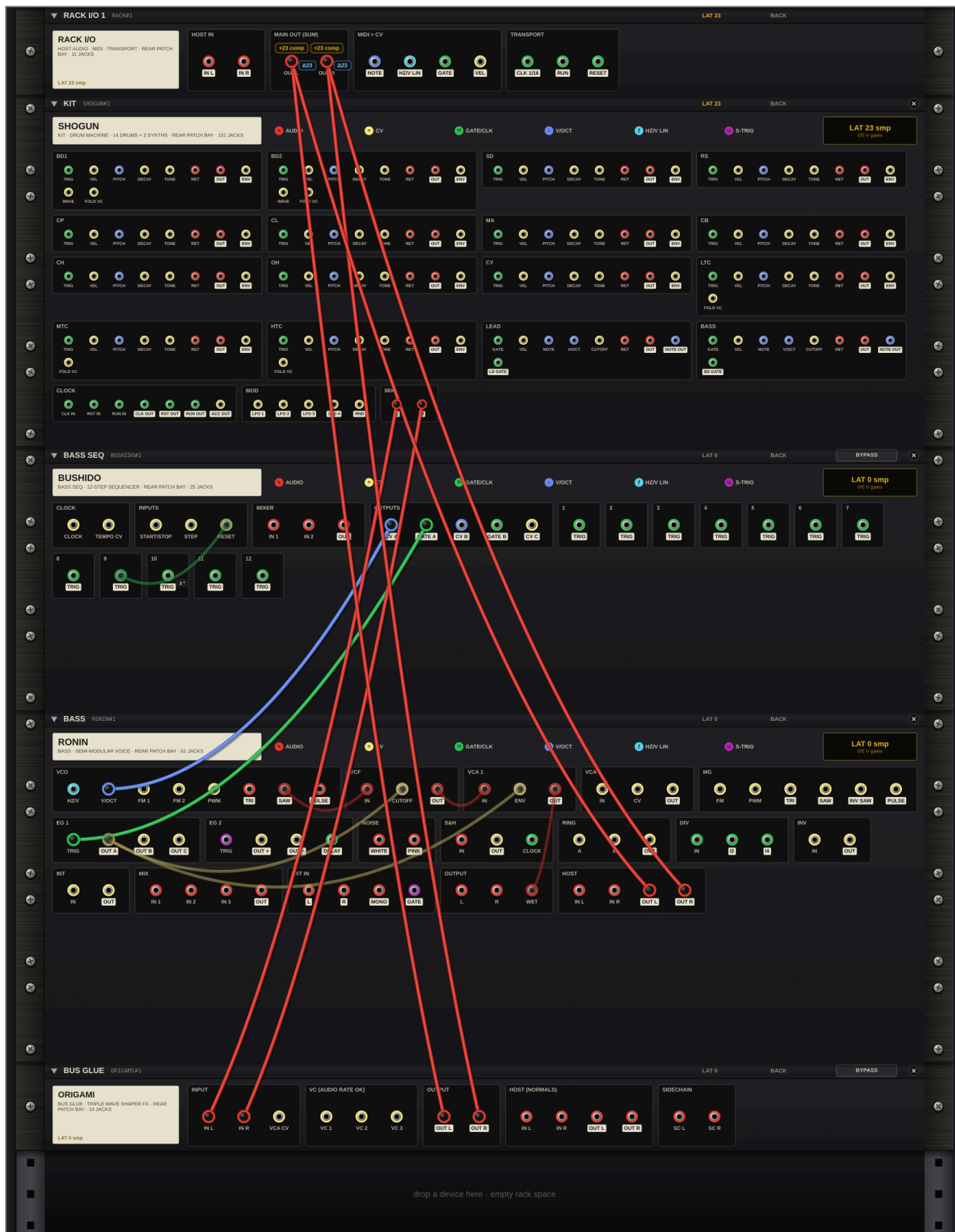


From	To	Why
RACK I/O › CLK 1/16	SHOGUN › CLOCK CLK IN	SHOGUN steps on every sixteenth note from the DAW.
RACK I/O › RESET	SHOGUN › CLOCK RST IN	Restarts SHOGUN's pattern when the DAW starts.
BUSHIDO › CV A, GATE A, CV C	RONIN	The acid line on RONIN.
RONIN › HOST OUT L	ORIGAMI › IN L	The bass through the folder.
ORIGAMI › OUT L/R	RACK I/O › MAIN OUT L/R	The folded bass to the DAW.
SHOGUN › MIX L/R	RACK I/O › MAIN OUT L/R	The drums to the DAW.

ORIGAMI at 2× delays the bass by 46 samples and SHOGUN at 2× delays the drums by 23, so the rack delays the drums by another 23 (the **+23 comp** tags on the SHOGUN cables). The bass reaches MAIN OUT next to SHOGUN's MIX, never through RET, so drums and bass arrive together.

Worked example: Full EDM Jam (starter rack)

RACKS ▼ › EDM › Full EDM Jam clocks SHOGUN from the host with no clock cables. SHOGUN's MIX L/R run through ORIGAMI to MAIN OUT, and BUSHIDO's bass on RONIN goes straight to MAIN OUT. SHOGUN at 2× delays the drums by 23 samples, so the rack delays the bass by 23 to match (LAT 23).



Try patching **SHOGUN > CLK OUT** into BUSHIDO's clock input with SHOGUN on SRC INT, so the drums set the tempo, or **SHOGUN > LFO 1** into an ORIGAMI VC input for a fold that moves with the groove.